PeerEval: Reliable Peer Evaluation among AI Models

Xinya Du, The University of Texas at Dallas
xinya.du@utdallas.edu louis.xyd@gmail.com xinyadu.github.io
Arena Academic Partnerships Program, Fall 2026 12 months $50,000

Research objective

We propose a peer-evaluation framework in which AI models review one another's answers and collectively assess their quality. Reviewers can share biases: Arena found a 37-percentage-point gap between three same-provider judges' preferences for their family and human votes [3]. We use mutual agreement to learn how much influence each reviewer should have within this framework. The central question is when peer evaluation produces reliable answer comparisons and model rankings despite shared biases.

Peer-evaluation framework and preliminary evidence

Models contribute answers and peer reviews. Reviewers compare anonymous answer pairs, and their votes are combined into answer-quality decisions and model rankings. The panel starts with equal reviewer weights. We score each reviewer's agreement with the weighted votes of reviewers from other model families, smooth and normalize these scores, and use them to update reviewer influence. The collective evaluation is then recomputed. This repeats until weights stabilize or reach an iteration limit. Human votes are held out to test the quality of the resulting peer evaluations.

Our prior peer-evaluation work [1] and PeerEval leaderboard and tool [2] provide the infrastructure for the project. A pilot covers 1,179 human-labeled battles with four judges. On 794 crowd battles, agreement weighting matched 67.8% of votes, versus 65.7% for equal weighting. These exploratory results motivate testing when agreement is useful evidence and when it repeats a shared bias.

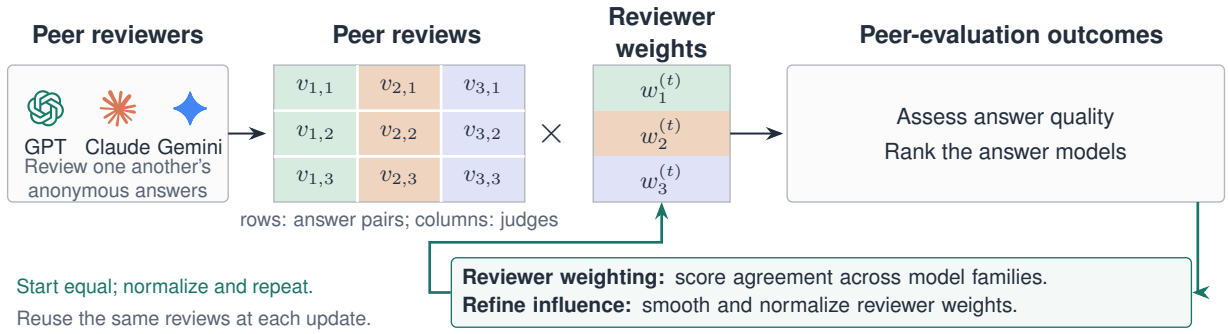


Figure 1: Peer evaluation among AI models. Peers review anonymous answers and combine their judgments into quality assessments and rankings. Mutual agreement with other model families updates each reviewer's influence within this process. Each iteration reuses the collected reviews. Icons illustrate model families.

Research approach

1. Learn reviewer influence within peer evaluation. Six to eight peer reviewers from different model families will compare anonymous answer pairs in both display orders, choosing A, B, tie, or both bad. We will align the orders and average each reviewer's votes. At iteration t, reviewer i receives an agreement score $a_i^{(t)}$: its mean agreement with the weighted votes of reviewers *outside its own model family*. This comparison excludes the reviewer's own vote and family.

Starting from equal weights, we update

$$w_i^{(t+1)} = (1-\eta)w_i^{(t)} + \eta \frac{\varepsilon + a_i^{(t)}}{\sum_j (\varepsilon + a_j^{(t)})},$$

where $0 < \eta \leq 1$ controls how quickly weights change and $\varepsilon > 0$ keeps every reviewer eligible. We stop when weights stabilize or reach a fixed iteration limit. The final weighted reviews determine answer preferences and model rankings. No human labels enter these updates.

2. Establish when peer evaluation is reliable. Agreement can reward a shared mistake, so we will study what prevents a group of similar judges from reinforcing itself. We will compare ordinary agreement, agreement excluding a judge's own family, and a version that limits any one family's total influence. We will analyze convergence, sensitivity to initialization, and how many correlated judges it takes to change a decision. Peer-assessed answer-quality weights [1] and a fixed combination of answer quality and agreement will serve as comparisons. These tests separate the benefit of the update rule from simply adding more judges.

3. Transfer peer evaluations to new questions and models. We will construct weights on development prompts, freeze them for unseen prompts, and compare this with recomputing weights on new unlabeled prompts. Later time windows and new answer-model families will test transfer. We will bootstrap questions and remove one judge family at a time to measure how stable the resulting rankings are. Instability will flag a result for additional review; we will test whether this flag predicts disagreement with people. For the proposed method, human votes are used only for evaluation.

Evaluation, outputs, and feasibility

We will use up to 10,000 public and, subject to agreement, recent Arena battles. Baselines include the strongest single judge, equal weighting, answer-quality weighting, static agreement weights, and weights fitted to limited human votes. We will measure preference accuracy, log loss with fixed probability smoothing, ranking agreement, stability, and cost, with paired bootstrap intervals that rerun the weighting loop. Length-matched tests, balanced longer/shorter-answer cases, and a length heuristic will check whether apparent gains come from verbosity. Repeated human votes and display-order metadata will help interpret disagreements. Shared biases may still limit gains; we will report those failure cases.

Project scope and requested support. Months 1–3 will establish data and baselines; months 4–8 will develop the peer-evaluation methods; months 9–12 will complete transfer tests and release a paper, code, and evaluation reports. The $50,000 gift allocates $36,000 to graduate research, including fringe and tuition; $10,000 to APIs and computing; and $4,000 to evaluation annotations. At maximum scale, 10,000 pairs $\times$ eight judges $\times$ two orders require 160,000 judging calls; pilot costs will set the affordable scale. Results will complement Arena's voting and AutoEval [4].

References

[1] Li R, Patel T, Du X. PRD: Peer Rank and Discussion Improve Large Language Model based Evaluations. TMLR, 2024. Paper.

[2] PeerEval. https://peereval-ai.github.io/; pilot results. 2026.

[3] Arena Team. Self-Preference of AI Judges. September 30, 2026. Article.

[4] Arena Team. Introducing AutoEval to the Arena leaderboards. July 30, 2026. Article.